COMSATS University Islamabad
Department of Computer Science

Course Information	
Course Code: **CSC354**	Course Title: **Machine Learning**
Credit Hours: **3(3,0)**	Lecture Hours/Week: **3**
Lab Hours/Week: **0**	Pre-Requisites: **None**

Text and Reference Books

Textbooks:

1. Introduction to Machine Learning, Ethem Alpaydin, MIT Press, 2010.
2. Machine Learning, Tom, M., McGraw Hill, 1997.

Reference Books:

1. Hands on Machine Learning with Scikit-Learn and TensorFlow, Aurelien Geron, O`Reilly Media, 2017.
2. Deep Learning with PyTorch – Essential Excerpts, Eli Stevens, Luca Antiga, Thomas Viehmann, Manning Publications, 2009.
3. Pattern Recognition and Machine Learning, Bishop, C., Springer-Verlag, 2007.

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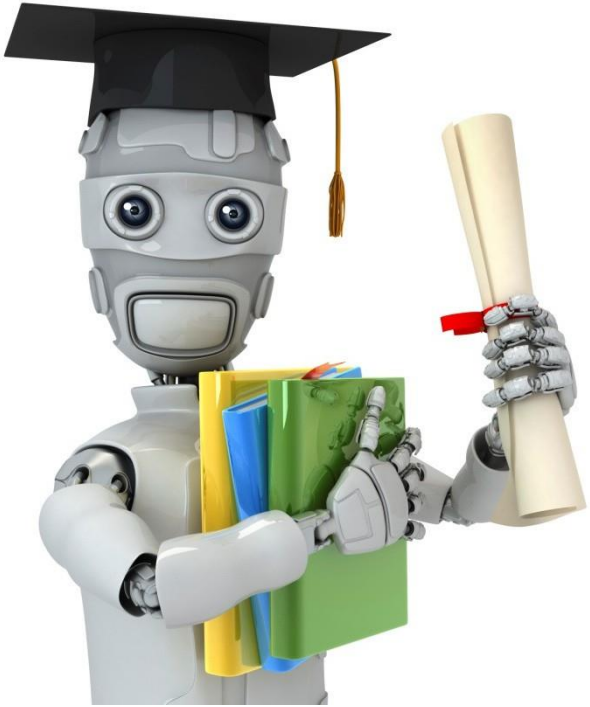
- Zero tolerance for any misconduct and misbehavior
- Zero tolerance for cheating, if student found with any cheating material in quiz/exam will be marked zero
- Copied assignment will be marked zero
- Cross talk with class-fellows will result in strict disciplinary action
- Be in class on time, students who arrive after 5 min will be marked absent
- Submit assignments on portal on time, late assignment wouldn't be considered, and anything send via (WhatsApp etc) wouldn't be considered
- Emails or any message should be sent within working hours rest would not be entertained send email on **(zeenatzulfiqar@cuiatd.edu.pk)**

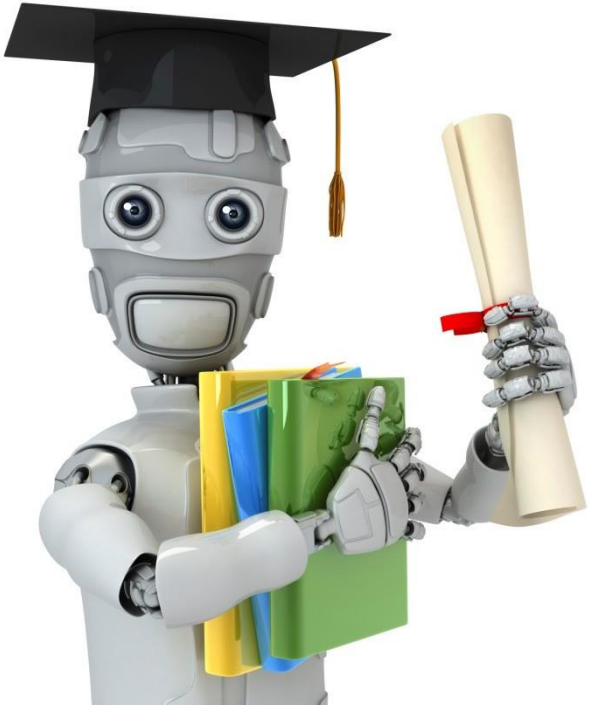


Machine Learning

Introduction

Contents

- What is machine learning
- Why Machine learning
- Applications of Machine learning
- Types of Machine Learning



Machine Learning

Introduction

What is machine learning

Machine Learning definition

- Arthur Samuel (1959). Machine Learning: Field of study that gives computers the ability to learn without being explicitly programmed.

- Machine Learning is the science (and art) of programming computers so they can learn from data.

Machine Learning definition

1. **“Learn from Data”**

- The system uses **data as input** (examples, records, history)
- It finds **patterns and relationships** in that data

Example:
An email system learns what is spam by analyzing thousands of emails.

Machine Learning definition

2. “Improve Automatically”

- The system gets better as it sees **more data**
- No need to rewrite code again and again

Example:
Netflix recommendations improve as you watch more movies

Machine Learning definition

Without Being Explicitly Programmed

- **Traditional Programming**

<u>In traditional systems:</u>

- You must write **every rule manually**
- The computer only follows your instructions

Example:

If email contains “free”, “win”, “prize” → mark as spam

If not → not spam

Problem:

Spammers change words (e.g., “fr33”, “w!n”)

You must **keep updating rules again and again**

Machine Learning definition

Without Being Explicitly Programmed

- **Machine Learning Approach**

<u>In Machine Learning:</u>

You **do NOT write rules**

You give:

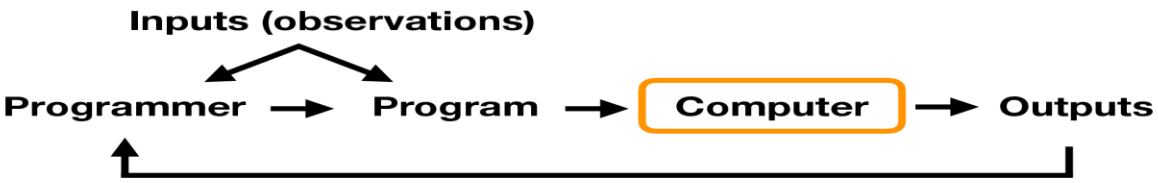
- Data (emails)
- Labels (spam / not spam)
- The system:
- Studies patterns in data
- Learns what features indicate spam
- Creates its **own rules automatically**

Machine Learning definition

“Make Intelligent Decisions”

- ML systems can **predict or decide outcomes**
- These decisions are based on learned patterns

Example:
- Predict house prices
- Detect fraud transactions



Machine learning is the field of study that gives computers the ability to learn without being explicitly programmed

— *Arthur Samuel (1959)*

Machine Learning definition

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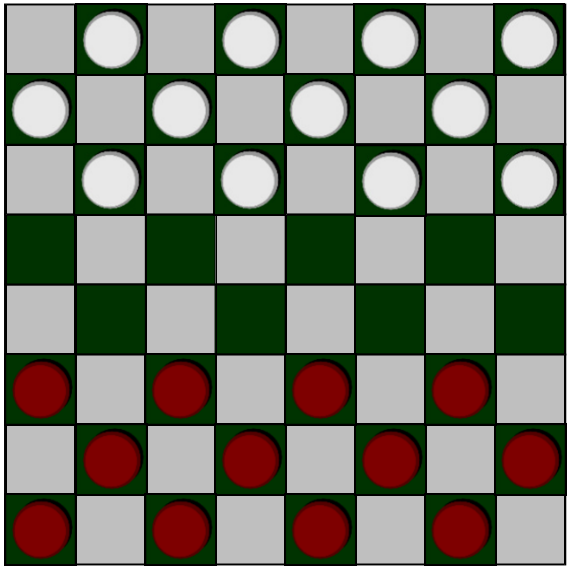


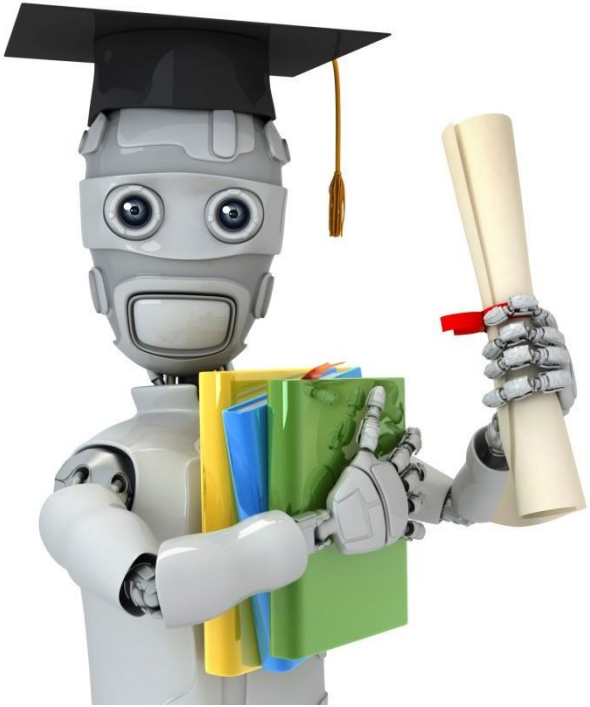
Machine Learning definition

- Arthur Samuel (1959). Machine Learning: Field of study that gives computers the ability to learn without being explicitly programmed.
- Tom Mitchell (1998) Well-posed Learning Problem: A computer program is said to *learn* from experience E with respect to some task T and some performance measure P, if its performance on T, as measured by P, improves with experience E.

"A computer program is said to *learn* from experience E with respect to some task T and some performance measure P, if its performance on T, as measured by P, improves with experience E."

Suppose your email program watches which emails you do or do not mark as spam, and based on that learns how to better filter spam. What is the task T in this setting?

- Classifying emails as spam or not spam. T
- Watching you label emails as spam or not spam. E
- The number (or fraction) of emails correctly classified as spam/not spam. P



Machine Learning

Introduction

Why Use Machine Learning?

Why Use Machine Learning?

1. Handling Complex Data

Today, we have massive amounts of data from:

•Social media

•Healthcare systems

•Sensors and IoT

Traditional programming cannot handle this efficiently.

Example:

A hospital has millions of patient records.

•Writing rules manually to detect disease patterns is impossible

•ML analyzes data and finds patterns automatically

Why Use Machine Learning?

2. No Need for Explicit Programming
In traditional programming:
You write exact rules
In Machine learning ML:
The system **learns rules from data**
Example (Spam Detection):
Traditional:
If email contains “win”, “free” → spam
Machine Learning:

- Learn from thousands of emails
- Automatically identify spam patterns
- More flexible and powerful

Why Use Machine Learning?

3. Improves Automatically (Learning from Experience)
ML systems:

- Learn from past data
- Improve performance over time

Example:
Netflix recommendation system:
At first: random suggestions
Later: suggests movies based on your history

Why Use Machine Learning?

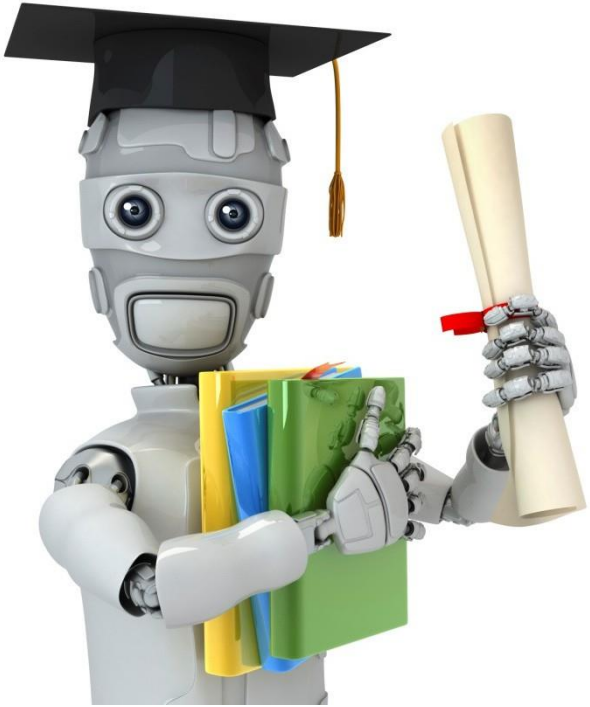
4. Solving Complex Problems

Some problems are too complex for manual coding:

- Image recognition
- Speech recognition
- Language translation

Example:

- Face recognition system:
- Instead of writing rules for eyes, nose, etc.
- ML learns from thousands of images



Machine Learning

Introduction

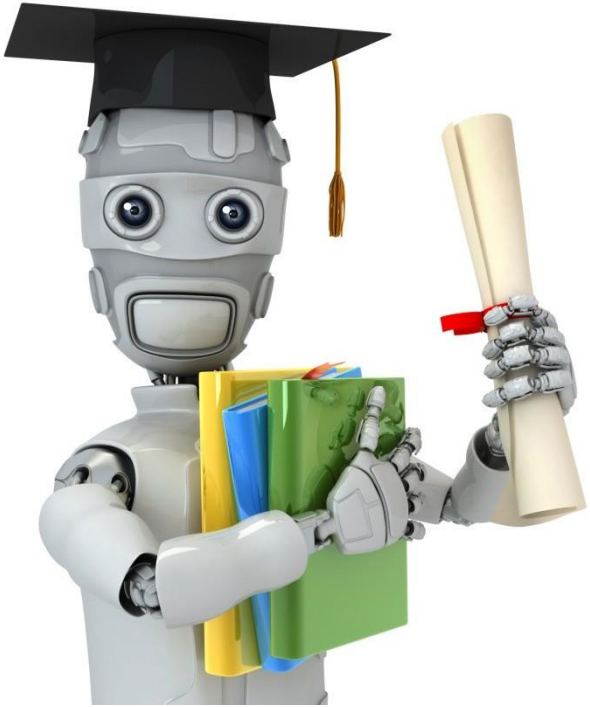
Types of Machine learning

Machine learning algorithms:

- Supervised learning
- Unsupervised learning

Others: Reinforcement learning

Also talk about: Practical advice for applying learning algorithms.



Machine Learning

Introduction

Supervised Learning

Supervised Learning

- **Supervised Learning (SL)** is a type of **machine learning** where the model is trained on a **labeled dataset**.
- "**Labeled dataset"** means every example in your training data has an **input** and a **correct output** (also called the target).
- The goal is for the model to **learn the relationship between input and output** so that it **can predict the output for new, unseen inputs**.

How Supervised Learning Works

The process can be divided into steps:

Step 1: Collect Data

You need a dataset with features (inputs) and labels (outputs).

Example: Predicting house prices

Input features: size of the house, number of bedrooms, location

Output/Label: price of the house

Step 2: Split the Data

Usually, we split the dataset into:

Training set: Used to teach the model

Test set: Used to evaluate how well the model performs on unseen data

Example: 80% training, 20% testing

How Supervised Learning Works

Step 3: Choose a Model
Some common models for supervised learning:

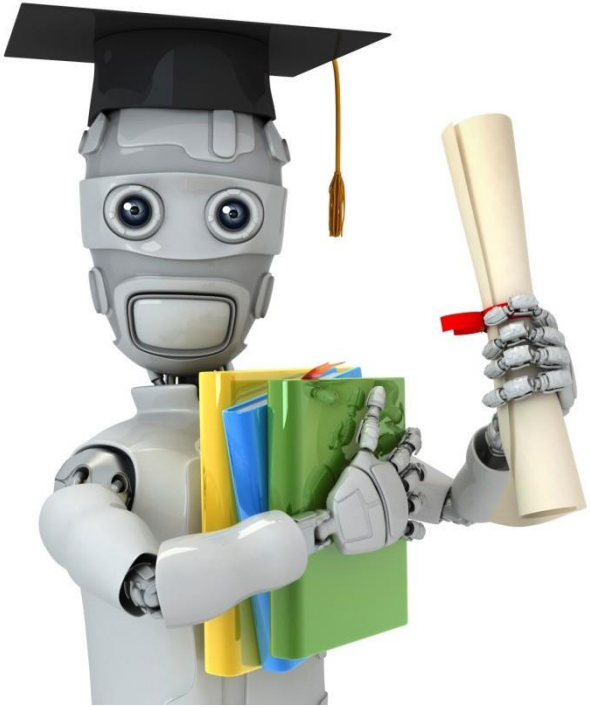
- **Linear Regression** → for predicting continuous values (like house prices)
- **Logistic Regression** → for predicting probabilities / categories
- **Decision Trees** → for classification or regression
- **Support Vector Machines (SVM)** → for classification
- **Neural Networks** → can do both classification and regression

Step 4: Train the Model
The model uses the training data to **learn the mapping between input and output**.
It adjusts its internal parameters to minimize the error between its predictions and the actual labels.
Example:
House with size 2000 sq ft → price 50,000$
Model predicts 48,000$ → error = 2,000$
Model adjusts parameters to reduce this error

How Supervised Learning Works

Step 5: Evaluate the Model
After training, test the model on the **test set**.
Common evaluation metrics:
Regression: Mean Squared Error (MSE), Root Mean Squared Error (RMSE)
Classification: Accuracy, Precision, Recall, F1-score

Step 6: Make Predictions
Once trained, you can feed **new input data** and the model will predict the output.
Example:
Input: house size 2500 sq ft, 3 bedrooms
Model predicts: price = 60,000$



Machine Learning

Introduction

Types of supervised Learning

Types of Supervised Learning

Supervised learning can be broadly divided into **two main types**:

1. **Regression**
2. **Classification**

Each type is used for a different kind of problem depending on the **type of output** we want the model to predict.

1. Classification

Classification is a type of supervised learning where the model predicts **discrete labels** or **categories**.

Discrete means the output can only be one of a **limited number of classes**.

The goal is to **assign each input to the correct category**.

Examples:

- **Email Spam Detection**

Input: Email text, sender, keywords

Output: Spam or Not Spam **(2 classes → binary classification)**

- **Digit Recognition (0-9)**

Input: Image of a handwritten digit

Output: Predicted digit **(10 classes → multi-class classification)**

- **Sentiment Analysis**

Input: Customer review text

Output: Positive, Negative, or Neutral **(3 classes → multi-class classification)**

2. Regression

- is a type of supervised learning where the model predicts a **continuous numeric value**.
- Continuous means the output can take **any real number** within a range, not just categories.
- The goal is to **find the relationship between input features and a numeric output**.

Examples:

House Price Prediction

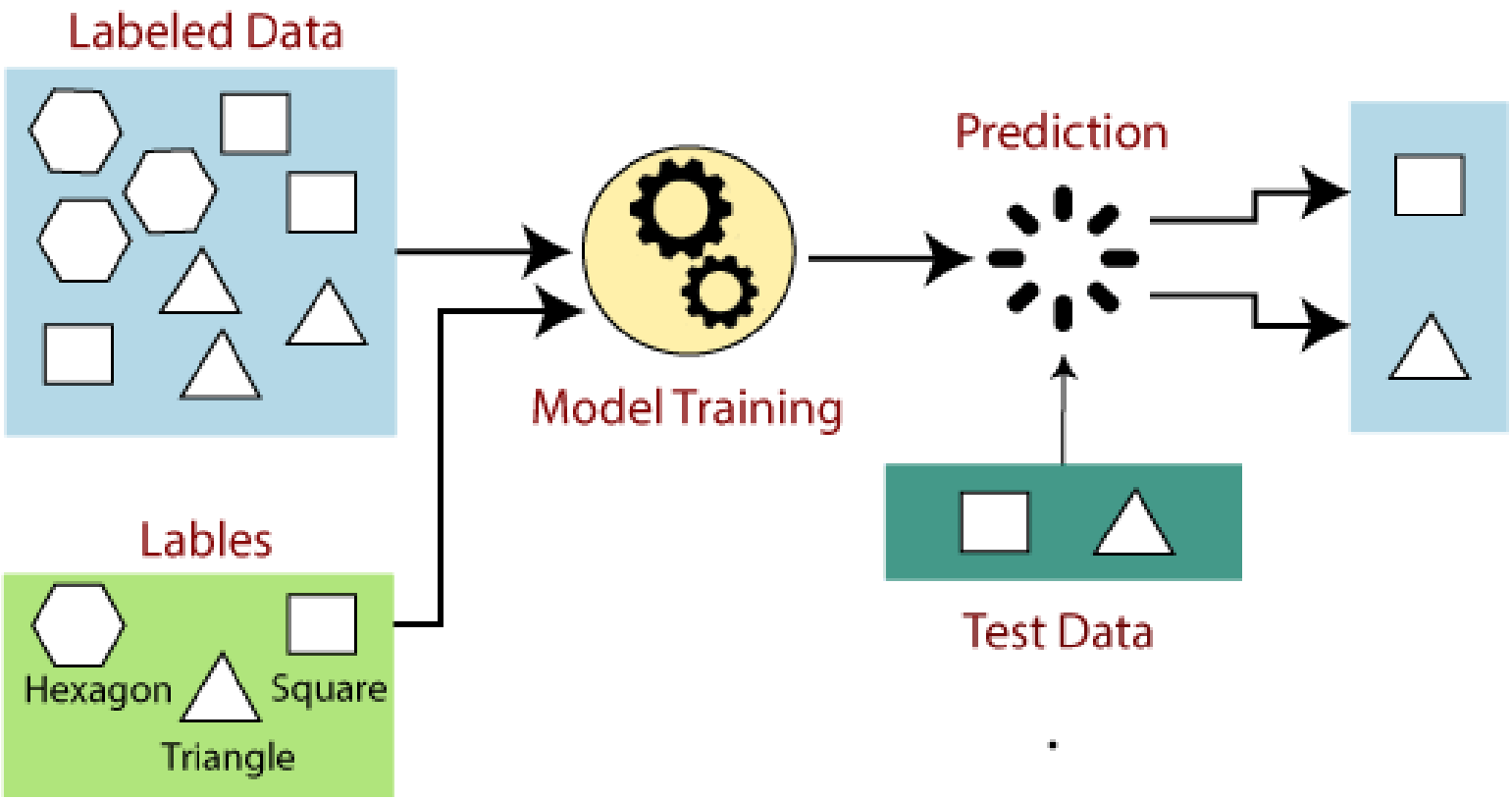
Input: Size of house, number of bedrooms, location

Output: Price in dollars

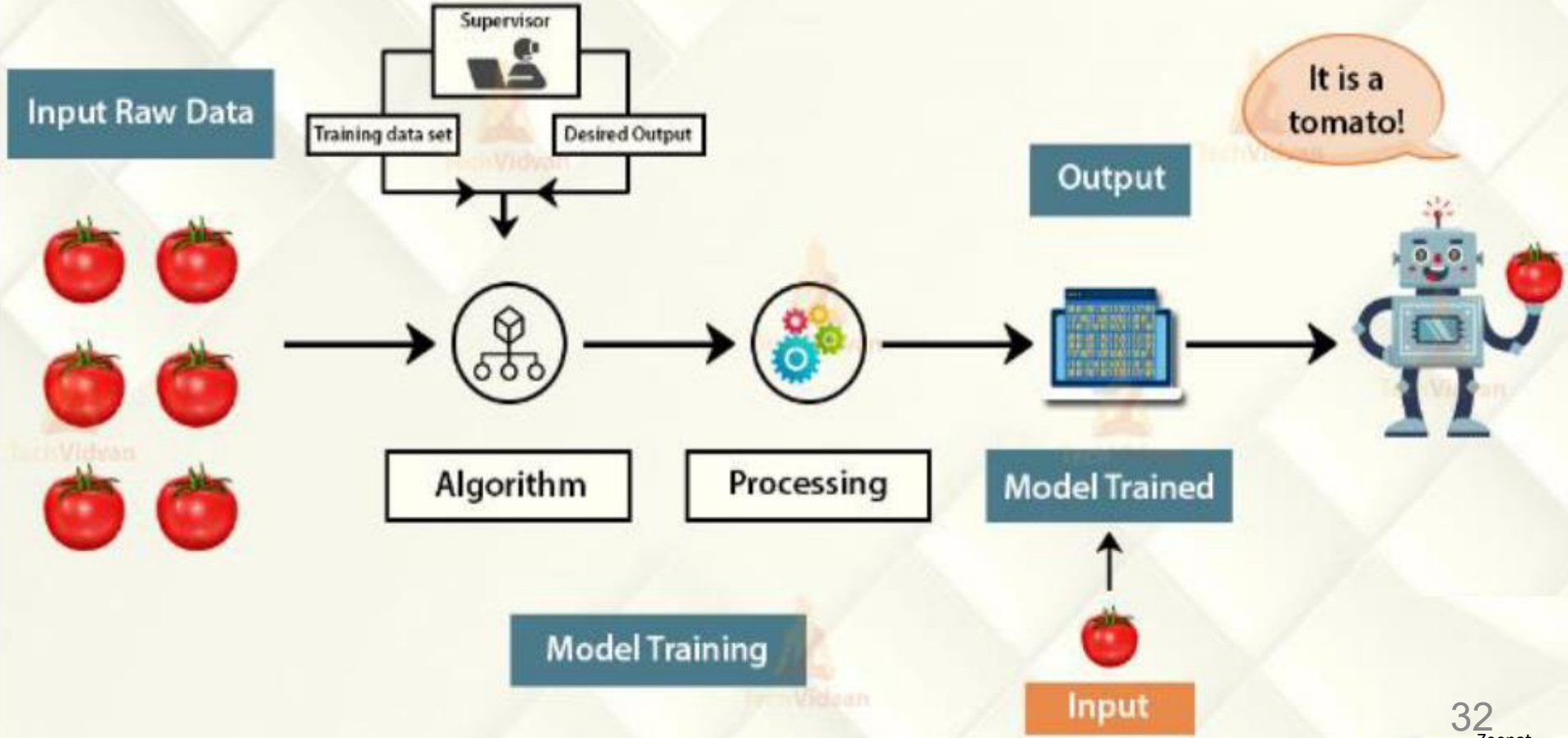
Stock Price Prediction

Input: Historical stock prices, market trends

Output: Predicted stock price


Labeled Data
Lables
Hexagon
Square
Triangle
Model Training
Prediction
Test Data

Supervised Learning in ML



Identify whether the task is **Classification** or **Regression**

S.No.	Scenario / Problem Statement	Input Features	Output (Target)	Type
1	Predicting House Prices	House size (sq ft), number of bedrooms, location	Price in USD	?
2	Email Spam Detection	Email text, sender, keywords	Spam / Not Spam	?
3	Weather Forecasting	Day of the year, humidity, atmospheric pressure	Temperature (°C)	?
4	Student Exam Performance	Hours studied, attendance, previous marks	Pass / Fail	?
5	Stock Price Prediction	Historical stock prices, trading volume, market trends	Predicted stock price (USD)	?
6	Heart Disease Diagnosis	Age, blood pressure, cholesterol level, lifestyle factors	Disease / No Disease	?
7	Vehicle Fuel Efficiency Prediction	Engine size, vehicle weight, fuel type	Miles per gallon (MPG)	?
8	Sentiment Analysis of Customer Reviews	Text of customer review	Positive / Negative / Neutral	?
9	Employee Salary Prediction	Education level, years of experience, job title	Salary in USD	?
10	Handwritten Digit Recognition	Image pixels of handwritten digit	Digit label (0–9)	?

S.No.	Scenario / Problem Statement	Input Features	Output (Target)	Type
1	Predicting House Prices	House size (sq ft), number of bedrooms, location	Price in USD	Regression
2	Email Spam Detection	Email text, sender, keywords	Spam / Not Spam	Classification
3	Weather Forecasting	Day of the year, humidity, atmospheric pressure	Temperature (°C)	Regression
4	Student Exam Performance	Hours studied, attendance, previous marks	Pass / Fail	Classification
5	Stock Price Prediction	Historical stock prices, trading volume, market trends	Predicted stock price (USD)	Regression
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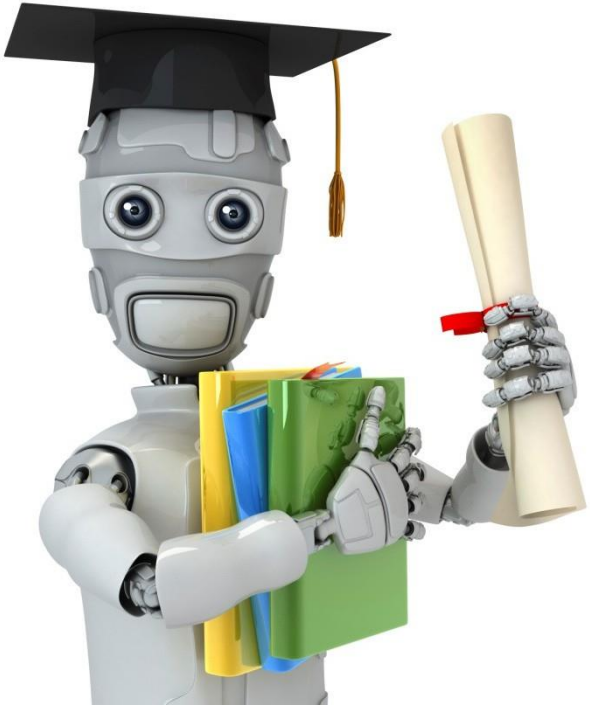
9	Employee Salary Prediction	Education level, years of experience, job title	Salary in USD	Regression
10	Handwritten Digit Recognition	Image pixels of handwritten digit	Digit label (0–9)	Classification



Machine Learning

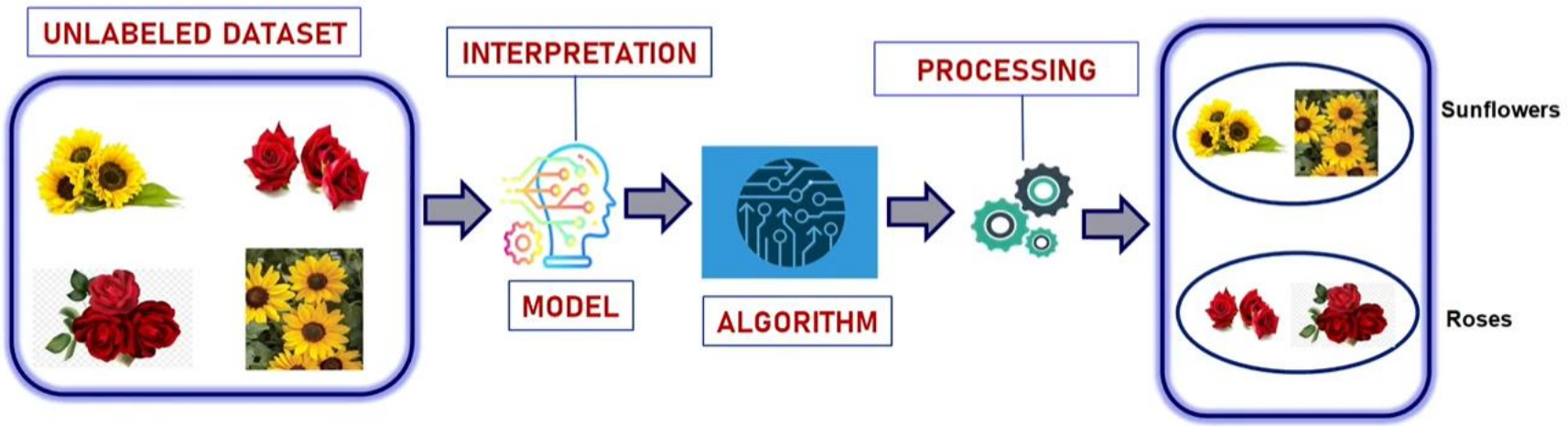
Introduction

Unsupervised Learning

Unsupervised Learning

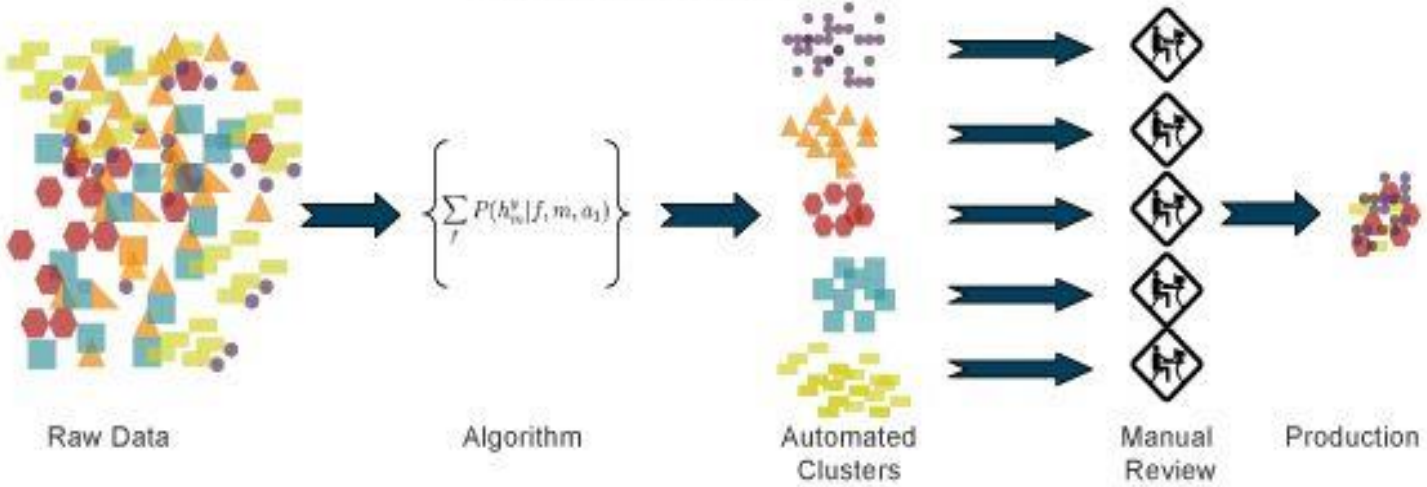
- Unsupervised Learning is a type of **machine learning** where the model learns patterns **from data that has no labels**.
- Unlike supervised learning, there is **no correct output or target** provided.
- The **goal** is to **find hidden structure, patterns, or groupings** in the data.
- Think of it like exploring a **new city without a map**: the algorithm tries to organize and understand the data on its own.

Key Points of Unsupervised Learning
No Labels: The dataset contains only inputs (features), **no outputs**.
Pattern Discovery: The model identifies **clusters, similarities,** anomalies, or underlying structure.
Goal: To **understand or summarize** the data rather than predict a specific output.


UNLABELED DATASET
INTERPRETATION
MODEL
ALGORITHM
PROCESSING
Sunflowers
Roses

UNSUPERVISED LEARNING

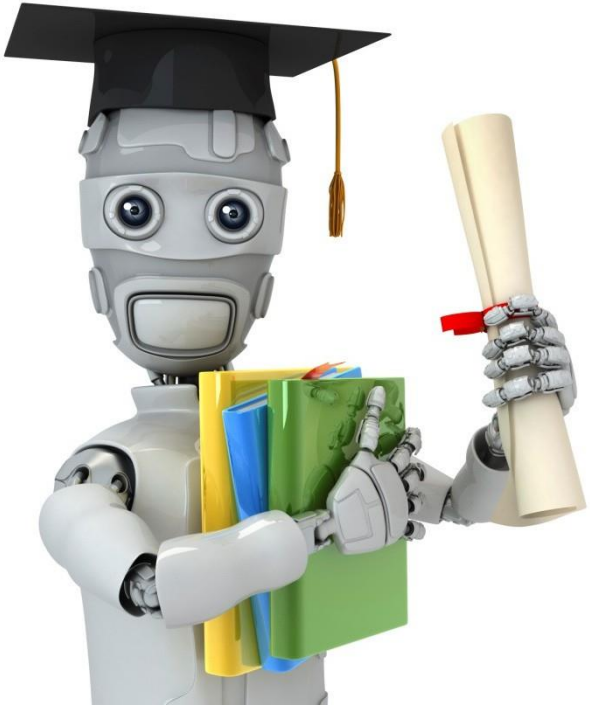
High reliance on algorithm for raw data, large expenditure on manual review for review for relevance and coding



E-Discovery Concepts: Machine Learning

Hudson | LEGAL

Zeenat



Machine Learning

Introduction

Types of Unsupervised Learning

1. Clustering

- Clustering is an **unsupervised learning** technique that **groups similar data points together** so that items in the same group **(cluster)** are **more similar to each other than to items in other clusters**.
- There are **no labels** in the **data;** the algorithm determines the groupings based on the **similarity of features.**

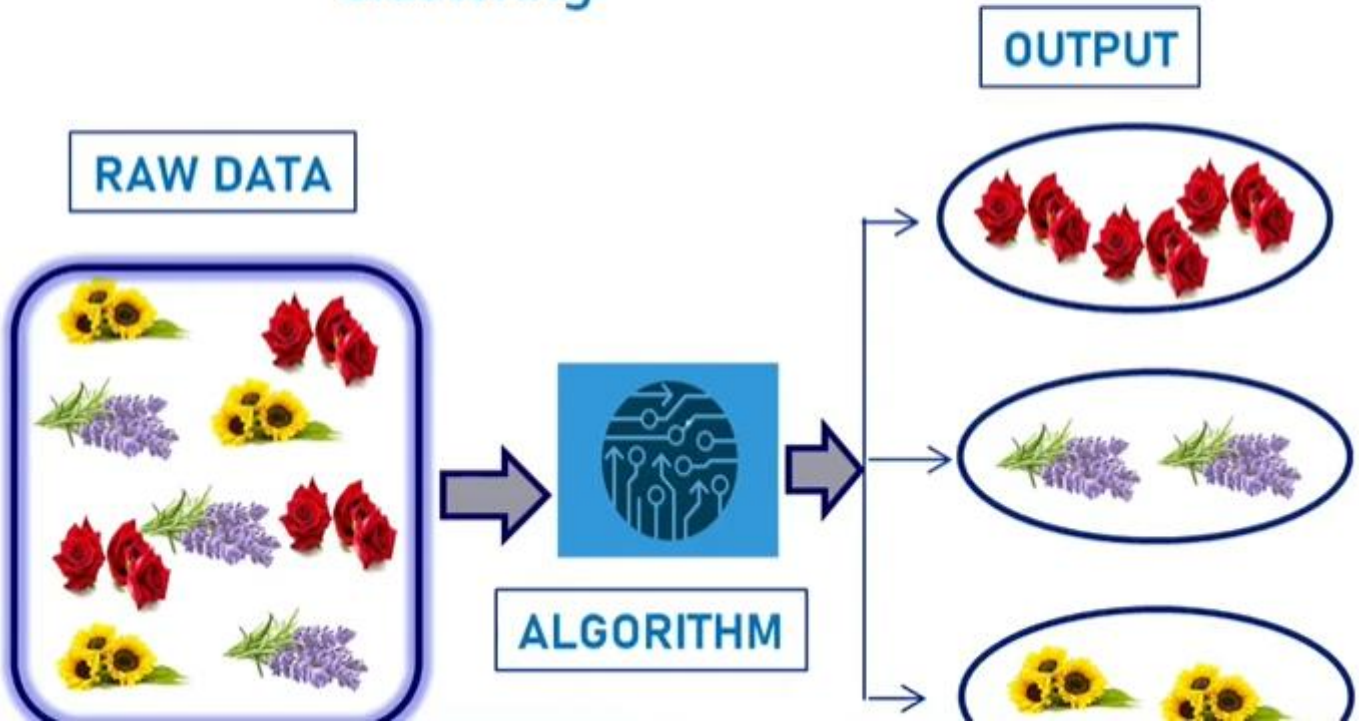
Goal of Clustering:

- Discover **hidden patterns or structure** in data.
- Organize data into meaningful groups for analysis.

Applications / Examples:

- **Customer Segmentation:**
- **Input:** Customer features like age, income, buying habits
- **Output:** Groups of customers with similar buying behavior

Clustering



2. Association

- **Unsupervised learning technique** used to **discover relationships or patterns among items in large datasets**.
- Unlike clustering, **association doesn't group data**; instead, it finds **rules like "If X happens, Y also happens"**.
- These rules show **how items co-occur**.

Goal of Association:

- Discover **frequent patterns, correlations, or dependencies** in data.
- Commonly used for **market basket analysis** in retail.

Applications / Examples:

- **Market Basket Analysis:**

Input: Customer purchase transactions

Output: Rules like:

"If a customer buys bread, they are 70% likely to buy butter"

Use: Cross-selling, product placement, promotions

- **Recommendation Systems:**

Input: User behavior on e-commerce websites

Output: Rules for product recommendations ("Customers who bought X also bought Y")

- **Medical Diagnosis:**

Input: Symptoms of patients

Output: Associations between symptoms and diseases

Customer 1

- *Bread*
- *Milk*
- *Fruits*
- *wheat*

Customer 2

- *Bread*
- *Milk*
- *Rice*
- *Butter*

Customer 3

Now, when customer 3 goes and buys bread, it is highly likely that he will also buy milk.

Identify whether the task is **Clustering** or **Association**

S.No.	Scenario / Problem Statement	Input Features	Output / Goal	Type
1	Group customers based on purchasing behavior	Customer age, income, purchase frequency	Customer segments / clusters	
2	Analyze supermarket transactions to find products frequently bought together	Transaction data: list of items purchased per customer	Rules like “If bread is bought → butter is also bought”	
3	Organize news articles by topic automatically	Words, keywords, frequency in articles	Clusters of similar articles	
4	Identify combinations of products to place together in a retail store	Sales data, product pairs in transactions	Rules like “If shampoo is bought → conditioner is bought 60% of the time”	
5	Segment patients based on medical test results	Age, blood pressure, cholesterol, sugar levels	Patient groups / clusters	
6	Find which movies are commonly watched together by users on a streaming platform	User watch history	Rules like “If user watches Movie A → likely watches Movie B”	
7	Group cities based on climate patterns	Temperature, rainfall, humidity	Clusters of cities with similar climates	
8	Market basket analysis in an online grocery store	Customer purchase lists	Rules like “If customer buys eggs → also buys milk”	
9	Segment students based on learning behavior	Attendance, grades, study hours	Groups of students with similar learning patterns	
10	Identify product bundles for promotion	Transaction data	Rules like “If customer buys phone → 70% buy phone case”	

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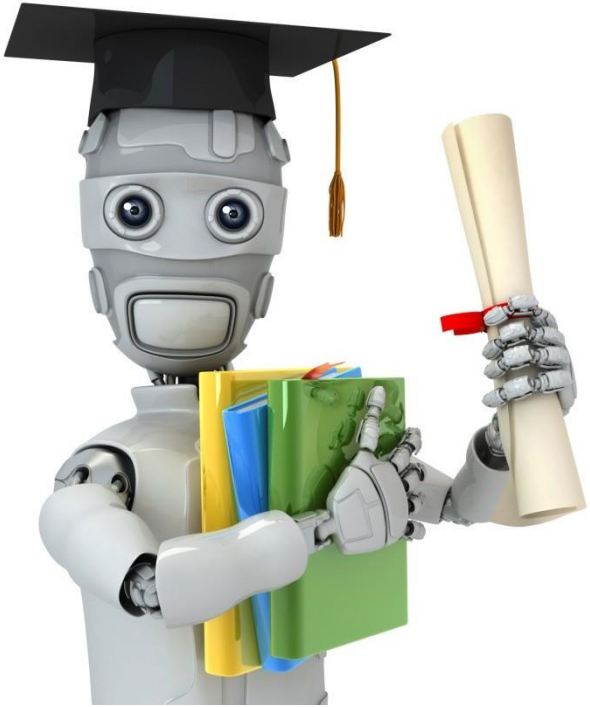


Machine Learning

Introduction

Reinforcement Learning

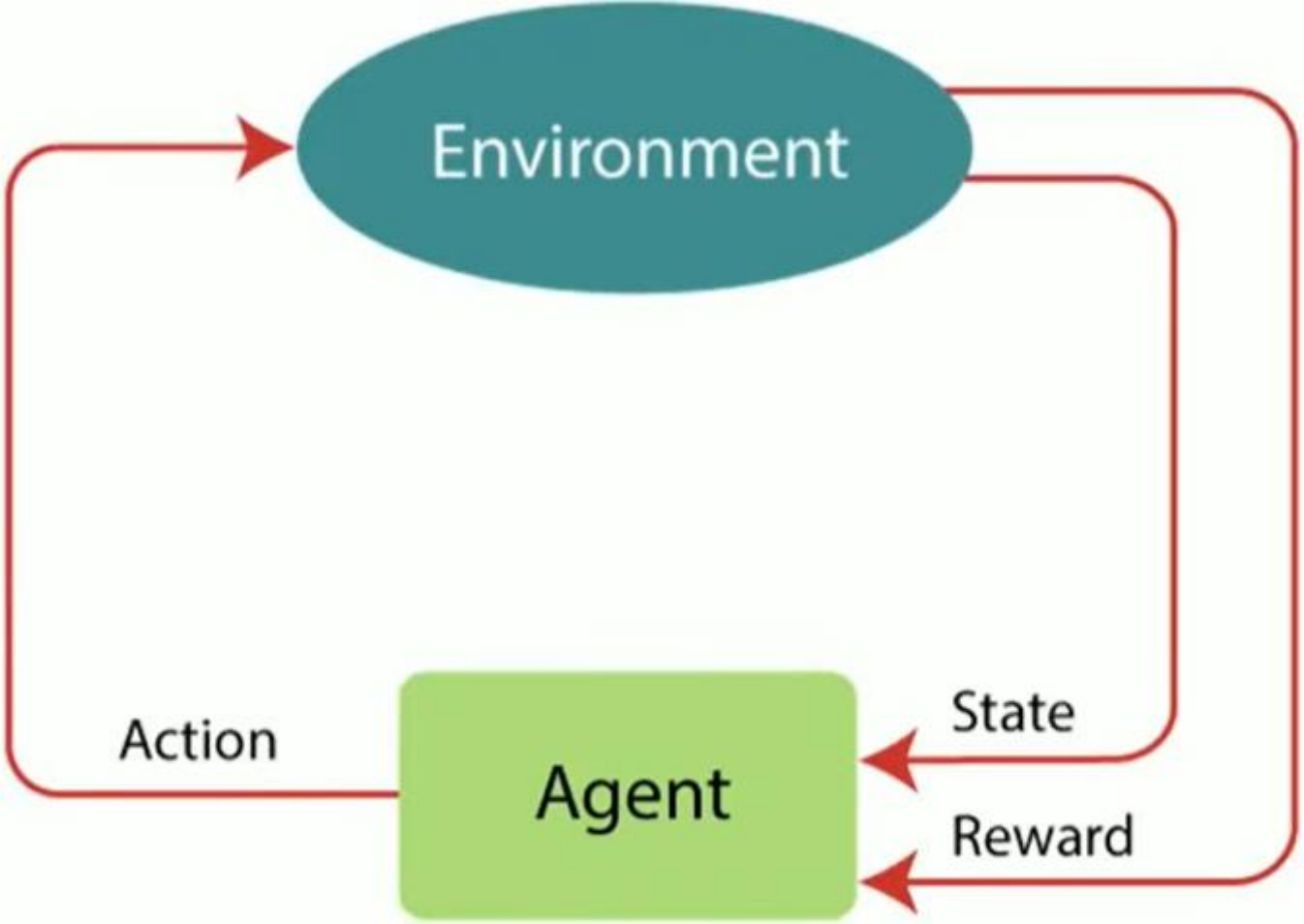
Reinforcement Learning

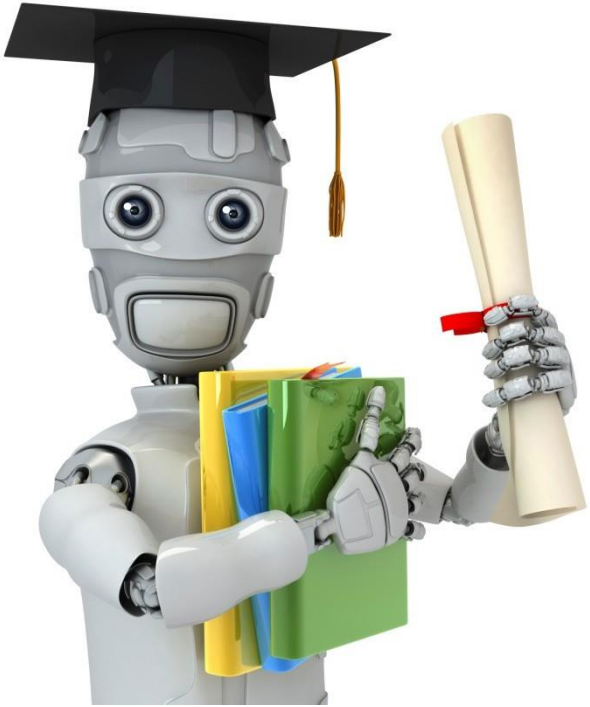
Reinforcement Learning is a branch of **machine learning** where an **agent learns by interacting with an environment** to **achieve a goal.**
Unlike **supervised learning**, RL **does not use labeled input-output data**.
Unlike **unsupervised learning**, it **does not just find patterns**.

Instead, it **learns optimal strategies through trial and error**.

Key Components

- **Agent:** The learner or decision-maker that takes actions
- **Environment:** The external system with which the agent interacts
- **Action:** The choices or steps taken by the agent
- **State:** The current situation or condition of the agent
- **Reward:** The feedback received from the environment based on the action


Environment
Action
Agent
State
Reward



Machine Learning

Introduction

Application of ML

MCQs on Reinforcement Learning

Reinforcement Learning is based on:
A) Labeled data
B) Trial and error
C) Supervision
D) Predefined rules

MCQs on Reinforcement Learning

In Reinforcement Learning, the learner is called:

A) Model

B) Agent

C) Dataset

D) Feature

MCQs on Reinforcement Learning

What does the agent receive after performing an action?

A) Label

B) Reward

C) Feature

D) Cluster

MCQs on Reinforcement Learning

Which of the following is an example of Reinforcement Learning?

A) Email spam detection

B) Image classification

C) Self-driving cars

D) Data clustering

Supervised, Unsupervised and Reinforcement learning

- **Scenario 1: Game Playing**
- A computer plays a game and improves after winning or losing.
- **Scenario 2: Email Spam Detection**
- Emails are labeled as spam or not spam.
- **Scenario 3: Customer Segmentation**
- Customers are grouped based on behavior without labels.
- **Scenario 4: Robot Learning**
- A robot learns to walk by falling and improving.
- **Scenario 5: Movie Recommendation**
- System improves suggestions based on user clicks.

Application: Disease Diagnosis & Prediction

Explanation:
ML models analyze medical data (X-rays, MRI scans, patient records) to **detect diseases** like **cancer, diabetes,** or **heart problems.**
Example: Predicting early signs of diabetes using patient data (age, BMI, blood pressure).
Benefit: Faster diagnosis, improved accuracy, reduced human error.

Application: Cybersecurity

Application: Threat Detection & Malware Analysis
Explanation:
ML detects **network anomalies, malware, or intrusion attempts**.
Example: Using ML to detect unusual login activity indicating a cyber attack.
Benefit: Protects data, prevents security breaches.

Finance & Banking

Application 1: Fraud Detection
Explanation:
ML detects **unusual transaction patterns** in real-time.
Example: Credit card fraud detection by analyzing spending behavior.
Application 2: Stock Market Prediction
Explanation:
Predict future stock prices based on historical trends using ML algorithms.
Benefit: Minimizes financial risks, improves investment decisions.

Energy & Utilities

Energy & Utilities
Application: Energy Consumption Forecasting
Explanation:
ML predicts electricity or gas usage based on historical data and weather conditions.
Example: Smart grids optimizing energy distribution.
Benefit: Efficient energy usage, cost savings, environmental benefits.

Robotics & Automation

Robotics & Automation
Application: Industrial and Service Robots
Explanation:
Robots use ML for **object detection, motion planning, and human interaction**.
Example: Warehouse robots sorting packages using computer vision.
Benefit: Increases productivity, reduces manual labor.

Education

Education
Application: Personalized Learning & Assessment
Explanation:
ML systems adapt learning material based on student performance.
Example: Online platforms suggesting lessons for students struggling in specific topics.
Benefit: Improves learning outcomes and engagement.

Supervised Learning Techniques

1. K-Nearest Neighbors (KNN)

Type: Can be used for **classification and regression**.

How it works:
•Looks at **K nearest points** in the training data.
•Classification: assigns the **most common class** among neighbors.

Example:
•Classification: Handwritten digit recognition.
•Regression: predicts the **average value** of neighbors.
•Regression: Predicting house price from nearby similar houses.

Supervised Learning Techniques

Applications of KNN

- **Recommendation Systems**: Suggests items like movies or products by finding users with similar preferences.
- **Spam Detection**: Identifies spam emails by comparing new emails to known spam and non-spam examples.
- **Customer Segmentation**: Groups customers by comparing their shopping behavior to others.
- **Speech Recognition**: Matches spoken words to known patterns to convert them into text.

Supervised Learning Techniques

Advantages of KNN

- **Simple to use**: Easy to understand and implement.
- **No training step**: No need to train as it just stores the data and uses it during prediction.
- **Few parameters**: Only needs to set the number of neighbors (k) and a distance method.
- **Versatile**: Works for both classification and regression problems.
- **Disadvantages of KNN**
- **Slow with large data**: Needs to compare every point during prediction.
- **Struggles with many features**: Accuracy drops when data has too many features.
- **Can Overfit**: It can overfit especially when the data is high-dimensional or not clean.